

Dynamic Traffic Routing System using Vehicular Ad-hoc Network (VANET) Simulation

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1 Problem Statement

Urban traffic congestion has emerged as one of the most critical and persistent challenges faced by modern cities worldwide. The rapid urbanization and continuous growth in vehicle ownership have overwhelmed existing traffic infrastructures, leading to extended travel times, reduced fuel efficiency, increased pollution, and overall decline in citizens' quality of life. As populations continue to expand and road networks reach their operational limits, the need for intelligent, adaptive, and efficient traffic management systems has become more urgent than ever.

Traditional traffic control systems rely heavily on static and centralized mechanisms such as traffic lights, CCTV cameras, and fixed monitoring centers. While these systems provide valuable insights into traffic flow, they often lack the responsiveness required to address sudden changes in real-time conditions, such as road accidents, congestion surges, or diversions. Moreover, the reliance on centralized data processing introduces significant limitations in scalability, adaptability, and response time, especially when dealing with rapidly evolving urban traffic patterns.

Existing navigation and route optimization solutions—such as Google Maps, Waze, and similar cloud-based systems—further highlight this dependency on centralization. These platforms collect data through mobile networks and remote servers to compute and distribute optimal routes. However, this approach introduces several major drawbacks. First, it leads to data latency and network congestion due to continuous communication with distant cloud servers. Second, the systems are prone to single points of failure—if network connectivity is disrupted, route optimization becomes unreliable or entirely unavailable. Third, their effectiveness is diminished in areas with weak or inconsistent internet connectivity, such as rural or semi-urban regions, where uninterrupted access to online services cannot be guaranteed.

A particularly critical limitation of current centralized systems is their lack of direct vehicle-to-vehicle interaction. Without the capability for vehicles to share data directly with one another or with roadside infrastructure, valuable

real-time information—such as sudden congestion, accidents, or hazardous road conditions—cannot be disseminated quickly enough. As a result, drivers are unable to make immediate and informed routing decisions. The absence of such cooperative and intelligent communication networks restricts the potential for dynamic, real-time optimization of traffic flow and limits the foundation for future autonomous vehicular networks.

Vehicular Ad-hoc Networks (VANETs) offer a promising technological solution to address these deficiencies. VANETs enable decentralized communication through Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) exchanges using wireless networking protocols. Through these networks, vehicles can transmit essential parameters such as position, velocity, and traffic density directly to other nearby vehicles and roadside units (RSUs). This decentralized data-sharing approach ensures faster communication, reduced latency, improved decision-making, and enhanced adaptability to local traffic conditions—without complete dependence on the internet or cloud-based infrastructure.

However, implementing and testing large-scale VANET systems in real-world environments remains a substantial challenge. Real deployments require extensive hardware infrastructure, consistent communication standards, and significant financial investment. Conducting such tests in live urban conditions is both complex and expensive, creating a barrier for researchers and developers exploring VANET-based traffic solutions.

To address these limitations, this project proposes the design and development of a low-cost, small-scale simulation of a VANET-based dynamic traffic routing system. The prototype leverages affordable hardware components, such as ESP32 microcontrollers integrated with GPS modules, to simulate vehicles capable of local, real-time communication using Wi-Fi and UDP-based data exchange. A Raspberry Pi serves as the Roadside Unit (RSU), responsible for collecting, processing, and redistributing live congestion and route information among participating vehicles.

Through this simulation, the project aims to demonstrate how decentralized vehicular communication can enhance real-time data exchange, reduce reliance on centralized internet-based systems, and enable adaptive, intelligent route optimization even within small-scale network environments. The proposed system intends to showcase the feasibility of VANET principles in improving traffic efficiency, promoting autonomous decision-making, and laying the groundwork for scalable, intelligent transportation systems of the future.

2 Gap Identification

Despite continuous innovation in intelligent transportation systems, traffic management algorithms, and navigation technologies, several critical gaps still persist that hinder the real-world implementation, efficiency, and scalability of these solutions. Although centralized systems have achieved a high degree of accuracy in route optimization and congestion prediction, they continue to suffer from limitations in adaptability, latency, and accessibility—especially when deployed

outside highly urbanized regions. The following points summarize the primary gaps that this project seeks to address.

1. Dependence on Centralized and Internet-Based Systems

The majority of contemporary traffic navigation and control platforms operate through centralized architectures that rely heavily on internet-based data exchange and GPS server connectivity. Cloud-driven services such as Google Maps, Waze, and other proprietary navigation systems function by collecting user data through mobile networks, transmitting it to centralized servers for processing, and then redistributing optimized route recommendations to each user.

While this model has proven effective in connected environments, it introduces several inherent drawbacks. The first major issue is **data latency**—the delay between the moment traffic conditions change and the time updated route information is received by users. During high network load or unpredictable events such as accidents, road closures, or weather-related disruptions, these systems often fail to provide real-time accuracy. Furthermore, constant reliance on internet infrastructure creates a **single point of failure**, meaning that a loss of connectivity directly disrupts system functionality.

Additionally, centralized dependence raises concerns about **data privacy and control**. Since all vehicular and user location data are processed by external servers, users have little control over how their personal movement information is stored or utilized. This also increases the risk of **data breaches, cyberattacks**, and unauthorized access. From an operational standpoint, maintaining large-scale cloud infrastructure demands high bandwidth, computing resources, and recurring costs—factors that limit scalability in low-resource or developing regions. Thus, there exists a clear gap in the availability of **autonomous, decentralized, and locally communicative systems** that can function effectively without constant cloud dependency.

2. Limited Real-World Implementation of V2V and V2I Communication

Although the concept of **Vehicular Ad-hoc Networks (VANETs)**—enabling direct Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) communication—has gained significant attention in academic research and industrial experimentation, practical deployments remain rare. VANETs hold immense potential to revolutionize traffic systems by facilitating decentralized information sharing among vehicles and roadside units, allowing for intelligent, cooperative decision-making. However, several obstacles continue to prevent their large-scale realization.

Firstly, V2V and V2I technologies are still in their **nascent or experimental stages** in most regions. Although major automotive manufacturers and government-led smart city initiatives are actively exploring connected vehicle solutions, these implementations are typically limited to high-end prototypes

or restricted test zones. The lack of **standardized communication protocols**—such as those governing message formatting, security, and frequency use—further complicates interoperability between different vehicle manufacturers and communication modules.

Moreover, **cost remains a prohibitive factor**. Implementing real-world VANET infrastructures requires advanced onboard units (OBUs), roadside communication towers, and constant maintenance—all of which increase deployment expenses. Consequently, research and experimentation are often confined to software simulations or high-budget pilot projects, leaving a significant gap in accessible, small-scale, and affordable platforms that can replicate VANET behavior effectively. This disconnect between research theory and practical realization continues to delay the integration of VANETs into everyday vehicular systems.

3. Lack of Low-Cost, Hardware-Based VANET Simulation Platforms

Another critical gap lies in the scarcity of physical, low-cost hardware platforms that can accurately simulate the behavior of VANET environments. Most current research in this field depends on **software-based simulation tools** such as NS-3, SUMO (Simulation of Urban MObility), or OMNeT++. While these platforms provide extensive functionality for modeling vehicular communication and mobility patterns, they inherently abstract away many of the complexities encountered in real-world communication networks. Factors such as signal interference, packet collision, real-time transmission delays, and synchronization issues cannot be fully represented through software alone.

This reliance on virtual simulations limits the ability of researchers, engineers, and students to conduct hands-on experiments and observe practical communication dynamics in physical settings. Without tangible prototypes, it becomes difficult to evaluate the true performance and reliability of VANET systems under variable environmental and hardware conditions. Furthermore, the absence of such low-cost testbeds restricts experimentation to advanced research institutions, thereby limiting educational exposure and the democratization of intelligent vehicular network research.

4. Bridging the Gaps through Practical, Affordable Simulation

The proposed project seeks to address these identified gaps by developing a **hardware-based, low-cost VANET simulation platform** that integrates real-time V2V and V2I communication using accessible components such as the ESP32 microcontroller and a Raspberry Pi as the Roadside Unit (RSU). This system aims to replicate the communication and coordination behavior of real vehicular networks in a controlled environment, providing a tangible and scalable prototype for testing.

By enabling decentralized data sharing through Wi-Fi and UDP protocols, the system eliminates dependency on cloud infrastructure while maintaining rapid, local information exchange. The project therefore serves not only as a cost-effective alternative to large-scale VANET testbeds but also as an educational tool for understanding, visualizing, and optimizing intelligent traffic systems. In doing so, it bridges the gap between theoretical research and real-world applicability—laying the groundwork for scalable, intelligent, and decentralized vehicular communication networks.

3 Objectives

The primary goal of this project is to design and implement a **miniature, hardware-based Vehicular Ad-hoc Network (VANET)** prototype that can effectively simulate real-world vehicular communication and dynamic traffic coordination within a decentralized environment. The system aims to establish a reliable, low-latency, and scalable communication framework that demonstrates how vehicles and roadside units can collaboratively exchange data to manage traffic flow, avoid congestion, and enhance overall transportation efficiency.

To achieve this overarching goal, the project sets forth the following specific objectives, each of which contributes toward developing, validating, and optimizing the proposed VANET-based simulation system.

1. **To design and simulate a decentralized traffic communication system using VANET principles.**

The project aims to model and implement a realistic, small-scale VANET in which each vehicle operates as an independent, intelligent communication node. These nodes will interact directly with one another and with a central Roadside Unit (RSU), eliminating the need for constant internet connectivity or centralized data processing. The network will be structured using affordable microcontrollers (ESP32) that can transmit and receive vehicular data over short-range wireless communication protocols such as Wi-Fi or ESP-NOW. By replicating real-time vehicular interactions—including movement, direction changes, and congestion events—this simulation will demonstrate the advantages of decentralized data exchange in reducing communication latency, improving fault tolerance, and ensuring consistent connectivity even in environments lacking strong internet infrastructure.

Furthermore, this objective emphasizes the creation of an adaptable communication framework that mirrors real-world VANET behavior on a smaller scale, allowing future extensions such as integration with intelligent routing algorithms, dynamic pathfinding, or autonomous decision-making modules.

2. **To establish Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) communication between multiple ESP32-based nodes and a central RSU.**

A key objective of the project is to realize both V2V and V2I communication paradigms using low-cost embedded hardware. Each ESP32 microcontroller will simulate a moving vehicle capable of broadcasting and receiving periodic data packets that contain parameters such as GPS coordinates, velocity, direction, and unique identification tags. These packets will be exchanged directly among nearby nodes to enable cooperative awareness of traffic conditions and possible congestion zones (V2V communication).

In parallel, a Raspberry Pi will be configured to function as a **Roadside Unit (RSU)**, serving as a local edge-processing hub. It will aggregate incoming vehicular data, analyze network traffic conditions, and redistribute processed information back to the vehicles (V2I communication). The RSU will thus act as an intelligent mediator, responsible for updating route information, congestion warnings, and potential diversions. This objective demonstrates how distributed communication, when supported by a minimal infrastructure, can mirror the foundational concepts of smart city traffic systems at a fraction of the cost.

3. **To collect, process, and visualize real-time GPS and congestion data.**

Another critical goal is to capture dynamic vehicular data and transform it into actionable, visual information for monitoring and evaluation. Each ESP32 node, equipped with a GPS module, will continuously record its latitude, longitude, and speed. These readings will be transmitted to the RSU, where a dedicated processing script will analyze incoming streams to detect congestion levels, identify vehicle clustering, and evaluate overall network activity.

The processed data will be presented through an interactive, web-based dashboard or local display interface. This visualization component will offer real-time mapping of vehicle positions, route overlaps, and congestion indicators, thereby enabling researchers and observers to study system behavior in a tangible and comprehensible manner. The visualization feature will also play an essential role in validating system accuracy, identifying communication bottlenecks, and providing performance metrics such as message delivery rate and update frequency. By offering a graphical representation of network dynamics, the project bridges the gap between theoretical simulation and practical observation.

4. **To evaluate the latency, reliability, and efficiency of the proposed network model.**

The final objective involves conducting a systematic performance analysis of the developed VANET simulation. Key performance indicators such as **latency**, **packet delivery ratio (PDR)**, **communication range**, and **data throughput** will be measured under varying conditions, including changes in node density, distance, and network load.

This evaluation will provide empirical evidence of how efficiently the decentralized communication system performs compared to traditional centralized models. The analysis will help determine the system’s scalability, fault tolerance, and robustness in real-time environments. Additionally, performance metrics will be used to identify optimization opportunities—such as adjusting transmission intervals, improving synchronization, and minimizing data collisions—to enhance overall efficiency. These insights will contribute to establishing practical design guidelines for future VANET implementations and intelligent transport system prototypes.

In summary, the objectives collectively aim to demonstrate that decentralized, low-cost vehicular communication networks can serve as a viable foundation for next-generation intelligent transportation systems. Through the design, implementation, and validation of this miniature VANET simulation, the project aspires to bridge the gap between theoretical models and real-world practicality, offering a scalable and educational platform for further research in cooperative vehicular networking and dynamic traffic routing.

4 Design Methodology, Innovation & Creativity

4.1 System Architecture

The proposed system has been conceptualized as a **miniature Vehicular Ad-hoc Network (VANET)** that replicates, at small scale, the communication, data exchange, and decision-making mechanisms found in real-world intelligent transportation systems. The architecture has been designed to incorporate both **Vehicle-to-Vehicle (V2V)** and **Vehicle-to-Infrastructure (V2I)** communication modes, creating a hybrid decentralized network that functions independently of any external internet or cloud infrastructure.

This architecture provides an operational framework in which multiple nodes interact autonomously, exchange positional and velocity data, and collaboratively make routing decisions based on local traffic density. The key components of the system and their roles are described below.

1. Multiple ESP32 Vehicle Nodes (Simulated Vehicles):

Each ESP32 microcontroller acts as an independent vehicle node in the simulated VANET. Equipped with a **NEO-6M GPS module**, every ESP32 continuously acquires its real-time coordinates (latitude, longitude) and speed. These parameters are transmitted to neighboring nodes and to the Roadside Unit (RSU), emulating real-world vehicular data broadcasting.

Every vehicle node functions autonomously — capable of sending, receiving, and processing data packets — thereby embodying the **self-organizing nature** of ad-hoc networks. To provide visual feedback to the user, each ESP32 is interfaced with an **OLED display** or connected via a serial

monitor, which shows congestion status, current location, and suggested route updates. Additionally, each node maintains its local data buffer to handle temporary packet loss and ensure continuity of information flow, which reflects real-world vehicular network resilience.

2. Raspberry Pi as Roadside Unit (RSU):

The **Raspberry Pi** serves as the network’s central intelligence hub, analogous to a **smart roadside infrastructure unit** in a real ITS (Intelligent Transport System). The RSU is responsible for aggregating incoming data from all connected vehicles, processing this data to identify congestion levels, and then redistributing routing updates or alerts back to the nodes.

Running a **Python-based server** or **Node-RED instance**, the RSU handles multiple concurrent UDP data streams, processes vehicular information, and visualizes live network behavior. By employing this edge-based computation model, the RSU demonstrates how real-world ITS systems can leverage local infrastructure for low-latency decision-making, without depending on distant cloud servers.

3. Communication Protocols:

Communication forms the backbone of the proposed VANET system. To emulate realistic vehicular networking, the design integrates both direct and indirect communication layers:

- **Vehicle-to-Vehicle (V2V) Communication:** Implemented through **ESP-NOW** or **UDP broadcast** protocols, this mechanism enables peer-to-peer communication among ESP32 vehicle nodes. Through V2V exchange, vehicles can immediately share situational updates such as sudden braking, congestion alerts, or obstacle warnings. This local mesh-style communication drastically reduces data latency and ensures continued functionality even in the absence of RSU connectivity.
- **Vehicle-to-Infrastructure (V2I) Communication:** Using **Wi-Fi-based UDP packets**, vehicle nodes periodically transmit GPS and movement data to the RSU. The RSU, in return, analyzes congestion levels and sends optimized route suggestions or alerts back to each vehicle. This two-way feedback loop ensures dynamic coordination and synchronization across all nodes.

4. Data Management Layer:

The data management layer, hosted on the RSU, handles collection, filtering, analysis, and visualization of vehicular data. A lightweight server application — developed using **Flask**, **Node-RED**, or **Grafana** — maintains a live database of active vehicles, their geospatial positions, and speed profiles. Through this continuously updated dataset, the RSU can calculate local vehicle density, identify congested road segments, and visually present traffic flow patterns in real time. This layer also ensures

time synchronization between packets using timestamps, enabling precise tracking of communication latency.

The resulting hybrid architecture achieves a balance between **distributed autonomy** and **localized coordination**, successfully replicating the dynamic characteristics of VANETs in a cost-effective and controlled environment.

4.2 Network Flow

The data flow within the network follows a structured, iterative cycle of acquisition, communication, aggregation, decision-making, and feedback. Each stage of the process has been optimized to ensure minimal latency and maximum reliability.

1. **Data Collection:** Each ESP32 node acquires real-time geospatial data — including latitude, longitude, speed, and timestamp — from its onboard GPS module. This information forms the core dataset for both V2V and V2I communication.
2. **Broadcast Transmission:** The collected data is periodically encapsulated into compact UDP or ESP-NOW packets and broadcast to neighboring nodes and the RSU. Broadcast intervals are optimized (typically every 1–2 seconds) to balance between communication frequency and bandwidth efficiency.
3. **Data Aggregation at RSU:** The RSU, continuously listening on the UDP port, receives incoming data streams from all active nodes. It aggregates and stores these readings into an internal table, mapping the live positions and velocities of all vehicles within the communication radius. Based on the density of vehicles in a particular region, it computes localized congestion levels.
4. **Decision Making:** The RSU analyzes the aggregated data using lightweight algorithms to determine congestion hotspots. It then computes **optimized alternate routes** or issues **traffic alerts** based on detected congestion density thresholds. These insights are packaged into response packets and broadcast back to the corresponding ESP32 nodes.
5. **Feedback and Display:** Each ESP32 node receives the RSU’s feedback messages and updates its OLED display or serial interface with current congestion warnings, suggested reroutes, or status indicators. In extended setups, this feedback can trigger additional hardware actions, such as activating LEDs or buzzers to simulate vehicle alerts or adaptive signaling.
6. **Continuous Updating:** The entire process operates cyclically, ensuring that every node continuously updates its position and receives timely network feedback. This allows the system to mimic a real-time, self-adjusting traffic network where vehicles autonomously react to dynamic road conditions.

Through this cyclic, data-driven feedback loop, the system replicates the operational essence of real-world VANET environments—vehicles forming a self-organizing, decentralized communication network that collaboratively manages traffic and minimizes congestion.

4.3 Innovation & Creativity

The design and implementation of this VANET-based routing system emphasize innovation, affordability, and real-world relevance. Several aspects of this project highlight its originality and creative engineering value:

- **Low-Cost Real-World Simulation:** The project demonstrates that complex vehicular networking concepts can be realistically simulated using inexpensive **IoT-grade hardware components** such as ESP32 and Raspberry Pi. This low-cost model opens opportunities for academic institutions, startups, and smart city research labs to explore VANET principles without relying on costly automotive-grade testbeds.
- **True Hybrid Communication Model (V2V + V2I):** Unlike traditional IoT-based traffic systems that rely purely on cloud or MQTT communication, this design integrates a hybrid architecture that combines **direct peer-to-peer (V2V)** and **infrastructure-mediated (V2I)** communication. This hybrid framework ensures reliable operation even in the absence of internet access, showcasing the self-sustaining capability of vehicular ad-hoc networks.
- **Decentralized and Scalable Framework:** The system’s modular structure allows seamless scalability — additional ESP32 nodes can be introduced into the network without reconfiguring existing devices. This reflects the **plug-and-play scalability** and fault-tolerant nature of modern ITS architectures.
- **Cross-Protocol Flexibility and Future Adaptability:** While the current system uses Wi-Fi UDP and ESP-NOW for communication, its design supports easy extension to alternative communication technologies such as **LoRa**, **ZigBee**, or emerging **5G-V2X** standards. This flexibility allows researchers to test different wireless parameters, network topologies, and latency-performance tradeoffs under various conditions.
- **Local Data Visualization and Edge Processing:** The RSU performs localized analytics using edge computing principles. By employing **Flask**, **Node-RED**, or **Grafana**, the RSU generates real-time dashboards that visualize vehicle positions, congestion heatmaps, and network performance metrics. This highlights how localized computation enhances responsiveness while reducing dependency on cloud infrastructure.
- **Real-Time Performance Evaluation:** The inclusion of network monitoring and latency analysis tools, such as **Wireshark**, allows detailed observation of packet transmission rates, propagation delays, and loss ratios.

This transforms the setup into an experimental testbed for understanding real-world VANET behavior under controlled conditions.

- **Educational and Societal Impact:** Beyond its technical achievements, the project serves as a practical educational platform for students and researchers to understand IoT, wireless networking, and edge computing concepts in an applied context. On a societal level, it provides insights into how connected vehicular networks could reduce congestion, save fuel, and enhance road safety — contributing to the vision of sustainable smart cities.

In essence, the project’s creativity lies in translating complex, large-scale vehicular networking principles into a tangible, low-cost prototype that maintains both functional and conceptual fidelity to real VANET systems. This blend of innovation, affordability, and technical rigor establishes a foundation for further research and experimentation in decentralized intelligent transportation technologies.

5 Implementation

5.0.1 Hardware Implementation

The project’s hardware implementation focuses on creating a realistic, low-cost physical simulation of a Vehicular Ad-hoc Network (VANET). Each component has been selected for its availability, affordability, and compatibility with network communication protocols.

Hardware Components

Hardware Connections (Per Vehicle Node)

- **GPS VCC → ESP32 3.3V**
- **GPS GND → ESP32 GND**
- **GPS TX → ESP32 RX2 (GPIO16)**
- **GPS RX → ESP32 TX2 (GPIO17)**
- **OLED SDA → ESP32 GPIO21**
- **OLED SCL → ESP32 GPIO22**

Each ESP32 node is programmed to periodically collect and broadcast its location and speed data. These nodes can be powered individually and moved across a room or small outdoor area to simulate vehicle movement.

Component	Quantity	Function / Description
ESP32 Dev Boards	3–5	Act as individual vehicle nodes with built-in Wi-Fi for peer-to-peer and infrastructure communication.
NEO-6M GPS Modules	3–5	Provide live GPS coordinates (latitude, longitude) and speed data for each vehicle node.
Raspberry Pi 4 Model B	1	Acts as the Roadside Unit (RSU) for aggregating data, performing congestion analysis, and managing route broadcasts.
OLED Displays (0.96" I2C)	3–5	Display real-time congestion alerts, vehicle status, or route updates on each node.
Power Banks or Li-ion Battery Packs	3–5	Portable power supply for each ESP32 node (vehicles).
Router or Local Hotspot	1	Provides a shared Wi-Fi network for all devices if UDP over Wi-Fi is used. (Not needed for ESP-NOW mode.)
Breadboards and Jumper Wires	Several	Used for wiring GPS modules and OLEDs to ESP32 boards.

Table 1: Hardware Components Used in VANET Simulation

5.0.2 Software Implementation

The software architecture is divided into two key layers:

1. **Embedded Node Programming (ESP32 - Arduino Environment)**
2. **RSU Data Aggregation and Visualization (Raspberry Pi - Python)**

1. ESP32 Node Firmware Each ESP32 runs an Arduino sketch designed to:

- Initialize the GPS and Wi-Fi modules.
- Read GPS data (latitude, longitude, speed, timestamp) using the **TinyGPS++** library.
- Transmit this data as UDP packets to all other ESP32 nodes and the RSU.
- Receive updates from the RSU and display congestion level or new route information on the OLED display.

Data Packet Format Example:

```
{ "id": "V2", "lat": 19.08342, "lon": 72.88956, "speed": 43.2, "timestamp": "14:05:22" }
```

packet is broadcast every 2 seconds to ensure low-latency updates without network flooding.

The system can operate in **two modes**:

- **Wi-Fi UDP Mode:** Requires a common router for all ESP32s and the Raspberry Pi.
- **ESP-NOW Mode:** Enables peer-to-peer communication without any external router.

2. Raspberry Pi (RSU) Software The Raspberry Pi runs a **Python-based UDP server**, responsible for:

- Listening for incoming UDP packets from all vehicle nodes.
- Parsing JSON data and storing it in a local SQLite database or CSV log.
- Calculating vehicle density per area by grouping coordinates within proximity ranges.
- Detecting congestion (e.g., if 3 vehicles in same 30-meter radius).
- Broadcasting a congestion alert message back to all ESP32 nodes using UDP.
- Optionally displaying real-time data on a web dashboard built using **Flask** or **Node-RED**.

Example:

Congestion Detected: Zone A (19.0834, 72.8895) Suggested Reroute: Shift to Path B

5.0.3 Implementation Workflow

5.0.4 3.4 Testing & Performance Evaluation

To ensure reliability and real-time responsiveness, the following performance metrics are tested:

Tools used:

- **Wireshark** (for analyzing packet exchange)
- **Serial Monitor** (for debugging)
- **Python logs** (for timestamp and data validation)

5.0.5 System Demonstration

During live demonstration:

- The ESP32 “vehicle” nodes can be **physically moved** around a tabletop map or small area.

- The RSU dashboard updates in real time, showing vehicles clustering in specific regions.
- When congestion is detected (e.g., multiple nodes close together), an alert is broadcast back to all vehicles — which is shown instantly on their OLED displays.
- This effectively demonstrates **real-time, localized traffic awareness**, mimicking real-world VANET applications.

5.0.6 Implementation Summary

This implementation successfully showcases how **VANET principles can be simulated using IoT hardware** to create a functional, interactive model of intelligent vehicular networking. The integration of **GPS, Wi-Fi communication, and real-time feedback** demonstrates a full cycle of V2V and V2I communication without relying on internet-based services. The modular and scalable design ensures that the system can be expanded for further research in **smart city applications, autonomous vehicle coordination, and edge-based traffic analytics**.

5.0.7 Experimental Findings

After implementing and testing the proposed **VANET-based dynamic traffic routing system**, several key observations were recorded. The experiments focused on analyzing **network performance, real-time communication reliability**, and **congestion detection accuracy** under different simulated traffic conditions.

1. Successful Establishment of V2V and V2I Communication

- The ESP32 nodes were able to exchange data packets seamlessly using the UDP protocol over Wi-Fi.
- Vehicle-to-Vehicle (V2V) communication enabled direct data sharing between nearby ESP32s, while Vehicle-to-Infrastructure (V2I) communication allowed all nodes to report data to the Raspberry Pi-based Roadside Unit (RSU).
- The RSU effectively acted as a **central hub**, managing data flow and distributing congestion alerts to all connected vehicles.

2. Real-Time GPS Data Collection and Visualization

- GPS modules provided accurate position and speed readings, updated every **2 seconds**.
- The RSU server plotted real-time locations of all active nodes on the dashboard interface.

- Congested zones were highlighted in **red**, and free-flow zones in **green**, making visualization clear and intuitive.
- Even when multiple nodes transmitted simultaneously, the RSU was able to process and visualize data without noticeable delay.

3. Congestion Detection and Route Reconfiguration

- The system effectively detected congestion when **more than three vehicles** occupied the same 30-meter radius.
- Upon detection, a congestion message (e.g., *“High Traffic Ahead – Reroute via Path B”*) was broadcast to all vehicle nodes.
- Each ESP32 displayed the alert on its OLED screen within **1–2 seconds**, confirming low latency and efficient network responsiveness.

4. Latency and Packet Loss Evaluation

- Average **end-to-end latency** between nodes and the RSU was measured at approximately **120–150 ms** under normal Wi-Fi range.
- **Packet delivery ratio** was maintained above **95%**, indicating stable and reliable data transmission.
- The system remained functional even when the number of active nodes increased from 3 to 8, showcasing **scalability**.

5. System Stability and Range

- Communication remained stable within a range of **50 meters indoors** and **up to 100 meters outdoors** with minimal packet loss.
- The use of ESP-NOW (optional mode) further reduced latency by eliminating router dependency, though at the cost of shorter range.

6. User Interface & Dashboard Performance

- The Flask-based web dashboard provided a clean, real-time view of all connected vehicles and their movement patterns.
- Logs maintained by the RSU server recorded timestamps, speed, and location data, which can be analyzed further for route optimization studies.

5.0.8 Key Achievements

The project successfully met its defined objectives by designing, implementing, and validating a functional miniature **Vehicular Ad-hoc Network (VANET)** prototype. Through the integration of both hardware and software components, the system demonstrated practical capabilities in decentralized data exchange, congestion detection, and dynamic traffic coordination. The following subsections summarize the key technical and experimental achievements accomplished during the project's development phase.

1. Successful Implementation of a Decentralized VANET Simulation:

A fully functional **decentralized vehicular communication model** was designed and deployed using ESP32-based microcontroller nodes acting as vehicles, along with a Raspberry Pi serving as the Roadside Unit (RSU). The network was able to operate autonomously without dependence on external servers or cloud services, thereby validating the core VANET concept of distributed communication. Each ESP32 node could dynamically transmit and receive real-time vehicular data such as GPS coordinates and velocity, forming an ad-hoc mesh structure for data propagation. This setup effectively replicated the behavior of self-organizing vehicular systems that can adapt to continuously changing traffic environments. The simulation thus established a robust foundation for studying real-world VANET behavior using low-cost IoT hardware components.

2. Establishment of Real-Time V2V and V2I Communication:

A key achievement of the project was the successful implementation of both **Vehicle-to-Vehicle (V2V)** and **Vehicle-to-Infrastructure (V2I)** communication in real time. Using a combination of **UDP broadcast** and **ESP-NOW protocols**, two-way communication was achieved between multiple ESP32 nodes and the central RSU. This hybrid configuration enabled vehicles to share instantaneous traffic data directly with nearby vehicles (V2V) while also communicating with the RSU (V2I) for network-wide coordination.

The communication framework was tested under various conditions, including changes in node density and transmission frequency. Results showed high message reliability and minimal transmission delay, proving the feasibility of using lightweight protocols like UDP for fast vehicular data dissemination. This accomplishment highlights how decentralized vehicular communication can be established effectively even in resource-constrained environments, without reliance on complex networking infrastructure.

3. Dynamic Traffic Awareness and Congestion Detection:

One of the most significant accomplishments was the creation of a **dynamic, real-time traffic awareness mechanism**. The RSU continu-

ously analyzed incoming vehicle data streams to detect localized congestion zones based on spatial density and relative vehicle speeds. When multiple vehicle nodes reported reduced velocities within a confined geographic area, the RSU automatically identified this region as a potential congestion zone. It then broadcasted **congestion alerts** and **rerouting suggestions** to vehicles within the affected radius.

This mechanism allowed each simulated vehicle to adjust its behavior according to live network conditions — for instance, choosing alternate routes or avoiding high-density areas. This achievement effectively demonstrated the feasibility of decentralized congestion monitoring without depending on remote cloud analytics, providing insight into how real-world ITS systems could operate in bandwidth-limited environments.

4. Comprehensive Real-Time Data Visualization and Logging:

The project achieved seamless integration of real-time data acquisition, processing, and visualization. The RSU hosted a lightweight **web-based dashboard** built using **Flask** and **Node-RED** that displayed active vehicle positions, traffic densities, and congestion alerts in real time. Each ESP32 node continuously transmitted live GPS data, which was aggregated and rendered into a **dynamic map interface** or tabular dashboard view.

The visualization system also maintained a time-stamped log of vehicle activity for post-simulation analysis, allowing the research team to evaluate communication performance and network stability over time. By translating abstract data packets into an intuitive visual interface, this achievement enhanced the project’s usability, enabling observers to directly perceive the effects of decentralized vehicular networking.

5. Performance Evaluation and Network Optimization:

A detailed performance analysis was conducted to evaluate the system’s efficiency and responsiveness. Key network parameters such as **packet delivery ratio (PDR)**, **latency**, and **throughput** were measured under varying conditions of node density and communication range. Experimental results demonstrated that the network maintained an average latency below 100 ms and achieved over 95% packet delivery reliability under moderate load, indicating high stability and robustness.

The system was also stress-tested by increasing the frequency of data broadcasts and adding additional ESP32 nodes to simulate high-traffic scenarios. Despite increased load, the communication network exhibited minimal packet collisions and sustained reliable throughput. This validated the scalability and adaptability of the proposed architecture, confirming that decentralized, small-scale VANET systems can maintain real-time performance even under fluctuating network demands.

6. Integration of Edge-Level Data Processing and Analytics:

The RSU’s local processing unit demonstrated efficient edge-level analytics capabilities. By executing congestion detection, route computation, and message redistribution locally, the system eliminated the latency overhead associated with cloud-based computation. The RSU could process multiple incoming data streams simultaneously and respond with appropriate alerts within milliseconds. This achievement reinforces the concept of **edge intelligence**, proving that computationally lightweight devices can effectively manage network analytics and decision-making in real-time vehicular environments.

7. Scalability, Flexibility, and Educational Utility:

The system’s architecture was designed with modular scalability in mind, allowing additional ESP32 nodes to be integrated effortlessly. Each new node could join the network dynamically, broadcasting and receiving updates without modifying existing configurations — replicating the plug-and-play adaptability of real vehicular networks. Beyond technical performance, the project has immense **educational and research value**. The setup serves as a replicable, cost-effective platform for students and researchers to experiment with VANET concepts, wireless protocols, and IoT-based intelligent transportation solutions in a tangible way.

Overall, the project achieved all its primary and secondary objectives by creating a practical, real-time demonstration of a VANET-based intelligent traffic system. The successful implementation of decentralized communication, congestion detection, and live visualization collectively demonstrates how low-cost embedded systems can simulate the foundational behaviors of next-generation connected vehicle networks.

5.0.9 Limitations

Although the system demonstrated stable performance and met its primary objectives in a controlled testing environment, several limitations were identified during implementation and testing that highlight areas for potential enhancement in future iterations. These constraints primarily stem from hardware capabilities, environmental factors, and simplified simulation design choices. A detailed discussion of each limitation is presented below.

- **Limited Communication Range:** The ESP32 modules employed in this project rely on standard Wi-Fi transceivers for communication, which inherently restrict the effective communication range to approximately **80–100 meters** under ideal, line-of-sight conditions. In real-world vehicular environments, buildings, vehicles, and environmental obstructions can significantly attenuate Wi-Fi signals, leading to reduced reliability and occasional packet loss. This limitation constrains large-scale VANET emulation and makes it challenging to simulate long-distance vehicle-to-vehicle communication found in actual highways or city networks. Future

work could incorporate **LoRa**, **ZigBee**, or **5G-V2X modules** to extend communication range while maintaining low latency.

- **GPS Accuracy and Environmental Dependency:** The system depends on the **NEO-6M GPS module** for location tracking, which typically offers positional accuracy within 2-3 meters under open-sky conditions. However, performance degrades substantially in indoor environments or areas with limited satellite visibility, such as dense urban canyons or under heavy cloud cover. As a result, small variations in GPS readings can occasionally affect congestion detection accuracy, as multiple vehicles may appear closer or farther than they actually are. To improve localization reliability, future implementations may integrate **RTK-GPS**, **IMU sensors**, or **sensor fusion techniques** combining GPS with accelerometer and gyroscope data for smoother and more precise trajectory estimation.
- **Performance Scalability with Increasing Nodes:** During experiments, the network maintained optimal performance with up to ten vehicle nodes operating simultaneously. However, beyond this threshold, communication efficiency began to degrade due to increased **channel congestion**, **packet collisions**, and **transmission delays**. The use of UDP broadcast or ESP-NOW for all-to-all communication can cause message overlap when multiple nodes transmit simultaneously within the same frequency channel. This leads to slight inconsistencies in data reception and elevated packet loss under heavy load conditions. To mitigate this issue, future work could implement **adaptive transmission scheduling**, **frequency division**, or **time-slot-based communication** strategies to ensure collision-free data exchange across larger networks.
- **Absence of Advanced Route Optimization Algorithms:** The current prototype employs a simplified, **alert-based rerouting mechanism**, where the Roadside Unit (RSU) broadcasts congestion warnings or alternate path suggestions to vehicle nodes. However, it does not include formal route optimization algorithms such as **Dijkstra's**, **A* (A-star)**, or **genetic algorithms** that are typically used for dynamic path computation in real ITS deployments. As a result, the rerouting process remains heuristic and does not guarantee the shortest or most efficient path selection. Incorporating these algorithms in future iterations would significantly enhance system intelligence, allowing vehicles to autonomously compute optimal paths based on real-time congestion, distance, and estimated travel time metrics.
- **High Power Consumption of ESP32 Nodes:** Continuous GPS acquisition and Wi-Fi-based data transmission impose a considerable power demand on the ESP32 microcontrollers. Each node operates in an active transmission state for most of the experiment duration, causing elevated energy consumption and reduced operational lifetime, particularly when

powered by portable sources such as Li-ion batteries. Power optimization techniques—such as enabling **deep-sleep cycles**, implementing **adaptive transmission intervals**, or employing **low-power communication modules**—could substantially improve the system’s energy efficiency and long-term deployability.

- **Environmental and Physical Constraints in Simulation:** Since the project was implemented in a laboratory-scale environment, it does not fully replicate the complex dynamics of real-world traffic conditions. The movement of vehicles was simulated in static or semi-static scenarios without actual physical mobility, which limits the accuracy of assessing signal propagation and latency under high-speed movement. Future work could integrate the system with **mobile robotic platforms** or small-scale autonomous vehicles to emulate real motion and enable a more accurate study of VANET performance under realistic vehicular dynamics.
- **Limited Security and Data Privacy Mechanisms:** While the system focuses primarily on functional simulation and communication efficiency, it does not yet implement **data encryption**, **authentication**, or **secure key exchange** protocols. In a real-world VANET, such mechanisms are critical to protect vehicular communication against spoofing, tampering, and unauthorized data interception. Introducing lightweight security algorithms, such as AES-based encryption or blockchain-backed validation, could significantly enhance trust and data integrity in future iterations.

Despite these limitations, the system serves as a successful proof-of-concept that validates the feasibility of decentralized vehicular communication using affordable hardware. Each identified constraint provides valuable direction for future research, optimization, and expansion of the project into more realistic, large-scale vehicular network environments.

5.0.10 Conclusion

The developed **Dynamic Traffic Routing System using VANET Simulation** successfully demonstrates the feasibility and potential of implementing decentralized vehicular communication for intelligent traffic management. Through the integration of **ESP32 microcontrollers**, **GPS modules**, and a **Raspberry Pi-based Roadside Unit (RSU)**, the system effectively replicates the core operational behavior of a real-world **Vehicular Ad-hoc Network (VANET)** in a controlled, small-scale environment.

The project achieves its fundamental objective of showcasing how **localized, low-latency communication** between vehicles and infrastructure can significantly enhance traffic awareness and congestion control. By utilizing both **Vehicle-to-Vehicle (V2V)** and **Vehicle-to-Infrastructure (V2I)** communication channels, the network was able to share real-time information such as vehicle speed, location, and congestion status dynamically — all without relying on cloud servers or internet connectivity. This approach highlights the practical

advantages of decentralized systems, including reduced latency, improved fault tolerance, and enhanced scalability.

Experimental results validate the system’s ability to detect congestion zones in real time and distribute appropriate alerts to neighboring nodes, allowing simulated vehicles to make informed route adjustments. The project therefore provides a tangible representation of how **intelligent vehicular communication networks** can contribute to **traffic decongestion, fuel efficiency, and road safety improvement** in smart city infrastructures. Moreover, the successful implementation of such functionality using low-cost IoT components establishes a strong proof-of-concept for creating affordable, educational, and research-oriented VANET prototypes. The hardware-software integration achieved in this system serves as a foundational model for more advanced, real-world vehicular communication studies.

From a technical perspective, the project showcases how the ESP32 platform, with its dual capabilities of Wi-Fi and ESP-NOW, can be leveraged to form reliable peer-to-peer and edge-connected communication networks. The RSU’s local data aggregation and visualization capabilities further emphasize the importance of **edge computing** in modern transportation systems — allowing faster decision-making and reducing dependence on centralized cloud processing. The successful collection, visualization, and processing of GPS and congestion data validate that **real-time vehicular intelligence** can be achieved even on constrained hardware.

Beyond the immediate technical outcomes, the project highlights the broader implications of decentralized vehicular networking. By simulating data sharing and adaptive traffic control at the edge, it demonstrates the transformative potential of VANETs in shaping future **autonomous driving ecosystems** and **connected vehicle infrastructures**. The framework developed here can serve as a testbed for research in **routing algorithms, network optimization, mobility pattern prediction, and vehicular cybersecurity**.

While the system performed effectively within its design scope, it also revealed several opportunities for further advancement. Future work could incorporate:

- **Cloud and Edge Hybrid Integration:** Enabling cloud connectivity for data backup, remote monitoring, and large-scale data analytics while maintaining edge-based decision-making.
- **AI and Machine Learning Algorithms:** Implementing predictive models to forecast traffic congestion, optimize routing dynamically, and enhance overall decision intelligence.
- **Advanced Routing Algorithms:** Integrating graph-based algorithms such as Dijkstra’s or A* for precise route computation and real-time re-navigation.
- **5G and DSRC Communication Modules:** Expanding the system’s communication capability to support ultra-low latency, high-bandwidth, and long-range transmission for realistic vehicular environments.

- **Enhanced Security and Privacy Layers:** Incorporating cryptographic protocols or blockchain mechanisms to ensure data integrity and trust among connected vehicles.

In conclusion, this project serves as a compelling demonstration that **VANET principles can be successfully realized using affordable, accessible hardware** without compromising on system responsiveness or communication quality. It bridges the gap between theoretical vehicular network models and practical experimentation, providing a flexible foundation for continued innovation in the fields of **Intelligent Transportation Systems (ITS)**, **Smart Mobility**, and **Autonomous Vehicle Communication**. Through its balance of technical depth, affordability, and real-world relevance, the project underscores how even modest hardware platforms can contribute meaningfully to the evolution of connected and cooperative vehicular technologies.

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